

LSM3259 Creative Assignment

By: Ethan Wong

Project Pictures



Write-Up

The mushroom is modelled after *Mycena chlorophos*, an agaric Basidiomycota from the genus *Mycena* (Mycobank, 2026). The *Mycena chlorophos* is one of the three bioluminescent fungi species we have in Singapore. The model itself is made out of paper mache, and subsequently painted with white paint as a base coat and subsequently fluorescent paint. The paint is supposed to mimic the fact that the *Mycena chlorophos* glows in the dark. The mushrooms of the species are gilled; to mimic this, I cut some paper cards into triangles and glued them to the cap and the stem.

The rationale why I chose to highlight this fungi species specifically is to bring attention to the fact that, firstly, it is one of the newly recorded bioluminescent fungi in Singapore (Lee, 2022), and secondly, to showcase that Singapore's fungi scene is full of interesting individuals. This species is the brightest of the three bioluminescent species that can be found in Singapore. During the day, this fungus can be very difficult to identify as the fungi from the *Mycena* genus can appear very similar. However, at night, it is where it shows its true colours. Therefore, the only way to confirm if a *Mycena* sp. that has been spotted in the day is this particular species would be to return to the mushroom at night and observe if it has this bioluminescent characteristic, as no other species in the genus shares this trait. This species is typically solitary; a single fruiting body is typically spotted, although groups have been spotted to grow together on occasion. It is usually found growing on decomposing wood like many other fungi species (Lee, 2022).

The species itself is still relatively understudied, and the exact mechanisms of its bioluminescence are not yet fully understood. However, there are a few things that we do know about how it produces its light. Trans-p-hydroxycinnamic acid is likely one of the few chemicals within the mushroom that causes the activation of bioluminescent capabilities in the mushroom (Teranishi, 2016). At sufficient concentrations, it induces the mature parts of the mushroom to light up. To light up, the mushroom has specific light emitters such as riboflavin and flavin adenine dinucleotide disodium (Teranishi, 2016a). These compounds are what give the *Mycena chlorophos* its distinctive green colour in the dark.

References

Lee, S. (2022). The status of bioluminescent fungal species in Singapore. *NATURE in SINGAPORE*.

<https://doi.org/10.26107/NIS-2022-0124>

Mycobank. (2026). *Mycena chlorophos*. Mycobank.org.

<https://www.mycobank.org/page/Name%20details%20page/386081>

Teranishi, K. (2016a). Identification of possible light emitters in the gills of a bioluminescent

fungus *Mycena chlorophos*. *Luminescence*, 31(7), 1407–1413. <https://doi.org/10.1002/bio.3129>

Teranishi, K. (2016b). trans-p-Hydroxycinnamic acid as a bioluminescence-activating component in the pileus of the luminous fungus *Mycena chlorophos*. *Tetrahedron*, 72(5), 726–733.

<https://doi.org/10.1016/j.tet.2015.12.027>